

CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 03

Module: Quantum Chemistry

- Q1. Ionization energy (IE) of an one-electron atom is the minimum energy required to remove the electron from the lowest energy state of the atom. Using Bohr's theory, find the radius (in atomic unit a_0) of the 1st Bohr orbit, and ionization energy (in atomic unit E_h) for the following atoms/ions:

Properties	H	He ⁺	Li ²⁺	Be ³⁺	B ⁴⁺
r [a_0]					
IE [E_h]					

Interpret your results.

- Q2. Which of the following operator-function combinations would yield eigenvalue equation? Find the eigenvalue of the eigenfunction.

$$(i) \hat{O} = \frac{d^2}{dx^2} \quad \text{and} \quad f(x) = \cos\left(\frac{x}{4}\right) \qquad (ii) \hat{O} = \frac{d}{dx} \quad \text{and} \quad f(x) = e^{-3.4x^2}$$

$$(iii) \hat{O} = \frac{d}{dx} \quad \text{and} \quad f(x) = \cos(ax) \qquad (iv) \hat{O} = \frac{d}{dt} \quad \text{and} \quad f(t) = e^{i\omega t}$$

$$(v) \hat{O} = \frac{\partial}{\partial y} \quad \text{and} \quad f(x, y) = xy^3 \qquad (vi) \hat{O} = \frac{d^2}{dx^2} + 2\frac{d}{dx} + 3 \quad \text{and} \quad f(x) = e^{ax}$$

- Q3. Show that $\Psi(x) = e^{ikx} + e^{-ikx}$ is an eigenfunction of the Hamiltonian operator $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ for a free-particle in one-dimension. What is the eigenvalue? What is the observable here? Is $\Psi(x)$ also an eigenfunction of the linear momentum operator $\hat{p}_x$?

- Q4. Show that $f(x, y, z) = (\sin ax)(\sin by)(\sin cz)$ is an eigenfunction of the following operator (known as Laplacian operator):

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

- Q5. Indicate which of the following functions are “acceptable” wavefunctions. Give reasons.

$$(i) f(x) = x^2 + 1, \quad \text{where } x \text{ can have any value} \qquad (ii) f(x) = \pm\sqrt{x}, \quad \text{where } x \geq 0$$

$$(iii) f(x) = \frac{1}{\sqrt{2}} \sin\left(\frac{x}{2}\right), \quad \text{where } \left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right)$$

$$(iv) f(x) = e^{-x^2}, \quad \text{where } 0 \leq x < \infty$$